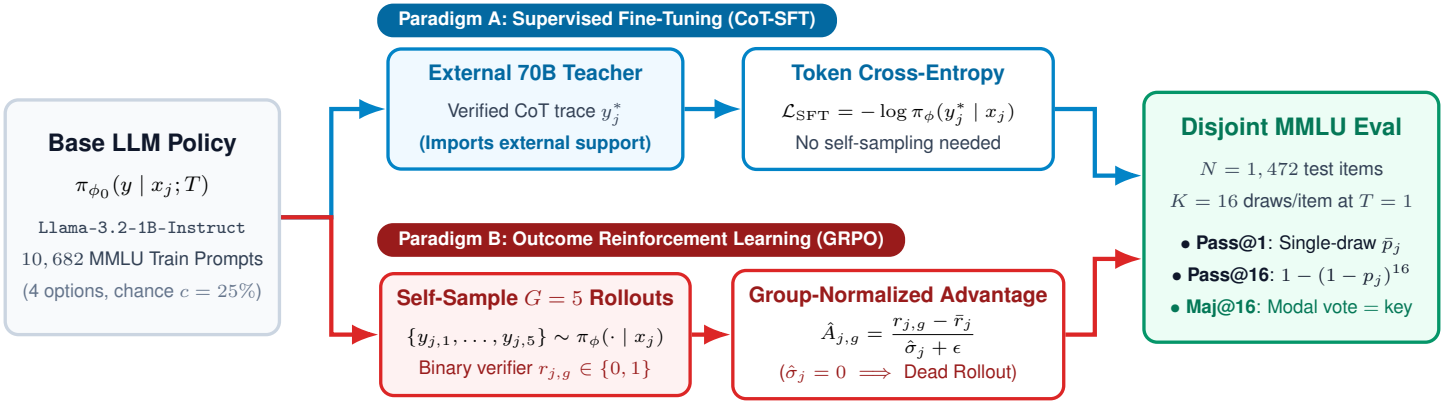

Base LLM Policy
$\pi_{\phi_0}(y \mid x_j; T)$
Llama-3.2-1B-Instruct
10,682 MMLU Train Prompts
(4 options, chance $c = 25\%$)
Paradigm A: Supervised Fine-Tuning (CoT-SFT)
External 70B Teacher
Verified CoT trace y_j^*
(Imports external support)
Token Cross-Entropy
$\mathcal{L}_{\mathrm{SFT}} = -\log \pi_\phi(y_j^* \mid x_j)$
No self-sampling needed
Paradigm B: Outcome Reinforcement Learning (GRPO)
Self-Sample $G = 5$ Rollouts
$\{y_{j,1}, \ldots, y_{j,5}\} \sim \pi_\phi(\cdot \mid x_j)$
Binary verifier $r_{j,g} \in \{0, 1\}$
Group-Normalized Advantage
$\hat{A}_{j,g} = \frac{r_{j,g} - \bar{r}_j}{\hat{\sigma}_j + \epsilon}$
($\hat{\sigma}_j = 0 \implies$ Dead Rollout)
Disjoint MMLU Eval
$N = 1,472$ test items
$K = 16$ draws/item at $T = 1$
• Pass@1: Single-draw $\bar{p}_j$
• Pass@16: $1 - (1 - p_j)^{16}$
• Maj@16: Modal vote = key